↓ Complex signal

Ex :1 $x(t) = e^{j\omega_0 t}$ $\forall\ t$ $\omega_0 t = \theta$

$e^{j\theta(t)} = \cos\theta(t) + j\sin\theta(t)$

$= \cos\omega_0 t + j\sin\omega_0 t$

$R = 1\Omega$

any signal $x(t)$

$x(t)$ is a real signal

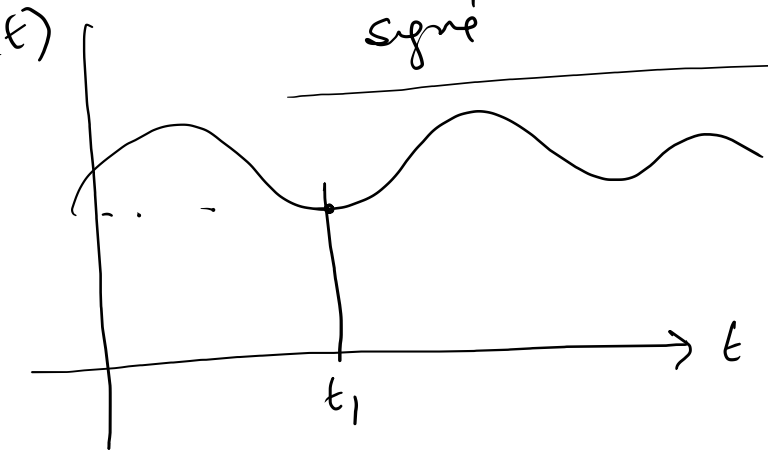
$$P_{av,\,t_2-t_1} = \frac{1}{t_2-t_1}\int_{t_1}^{t_2} x^2(t)\,dt$$

real signal

$x(t)$

if $x(t)$ is a complex signal $= P_{av,t_2-t_1} = \frac{1}{t_2-t_1}\int_{t_1}^{t_2} |x(t)|^2\,dt = \frac{1}{t_2-t_1}\int_{t_1}^{t_2} x(t)\,x^*(t)\,dt$

$x^*(t)$ = complex conjugate of $x(t)$



Ex

Cartesian coordinate system $x + iy$, $a + ib$

polar " " $re^{j\theta}$

Euler's identity $\Rightarrow$ $e^{j\theta} = \cos\theta + j\sin\theta$

	polar	Cartesian
(a)	$\frac{1}{2} e^{j\pi}$	$-\frac{1}{2}$
(b)	$\frac{1}{2} e^{-j\pi} = \frac{1}{2}\left[\cos(-\pi) + j\sin(-\pi)\right]$ $= \frac{1}{2}\left[\cos\pi - j\sin\pi\right]$ $= -\frac{1}{2}$	$-\frac{1}{2}$
(c)	$e^{j\pi/2} = \cos\pi/2 + j\sin\pi/2$	j
(d)	$\sqrt{2}\, e^{j9\pi/4} = \sqrt{2}\left(e^{j2\pi} \cdot e^{j\pi/4}\right)$ $\left(e^{j2\pi} \Rightarrow 1\right)$	$\sqrt{2}\, e^{j\pi/4} = \sqrt{2}\left[\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}\right] = 1 + j$

Convert from Cartesian to polar $\quad re^{j\theta}$ $\quad -\pi \le \theta \le \pi$

Ex $\quad 5 \longrightarrow 5e^{j0}$, ~~$5e^{j2\pi}$~~ $\quad \downarrow$

$\frac{1}{2} - j\frac{\sqrt{3}}{2} \longrightarrow re^{-j\theta} = e^{-j\pi/3}$ $\quad$ (+ve)

$-5 \longrightarrow 5e^{j\pi} = 5(\cos\pi + j\sin\pi) = -5$

r = magnitude

$\frac{1+j}{1-j} \longrightarrow \frac{(1+j)^2}{1^2 - j^2} = \frac{1 + j^2 + 2j}{2} = \frac{1-1+2j}{2}$

$= j = e^{j\pi/2}$ $\quad r = 1$, $\theta = \pi/2$

Energy $= E_\infty = \int_{-\infty}^{\infty} |x(t)|^2 \, dt$

$P_\infty = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 dt$

$$x(t)\Big|_{t=2} = a + ib$$

$$|x(t)|^2\Big|_{t=2} = a^2 + b^2$$

$$|x(t)|^2\Big|_{t=2} = x(t)\,x^*(t)\Big|_{t=2} = (a+ib)(a-ib)$$

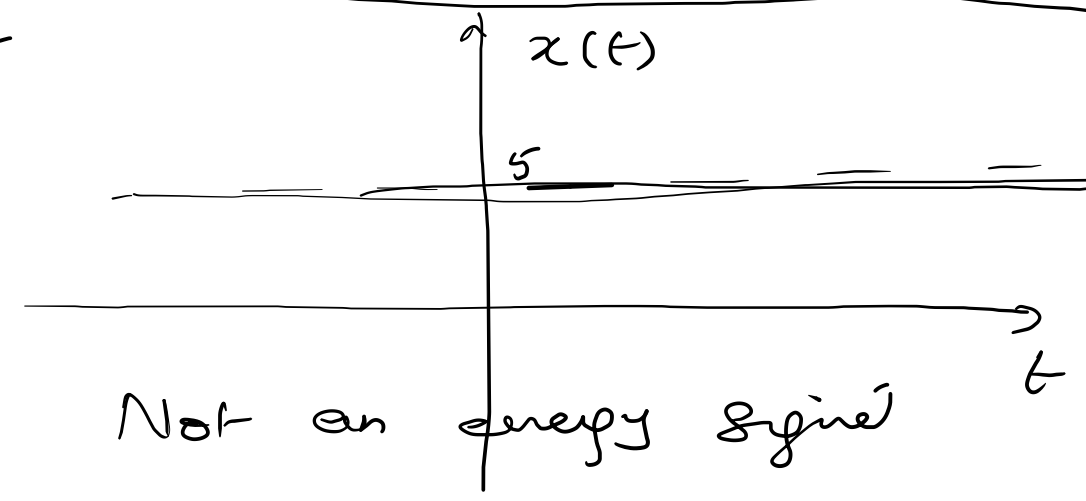
$$= a^2 + b^2$$

Ex-1

$$x(t) = 5 \quad \forall t$$

$$E_\infty = \int_{-\infty}^{\infty} |x(t)|^2\,dt = \int_{-\infty}^{\infty} 25\,dt$$

$$25\int_{-\infty}^{\infty} dt = \infty$$



Not an energy signal

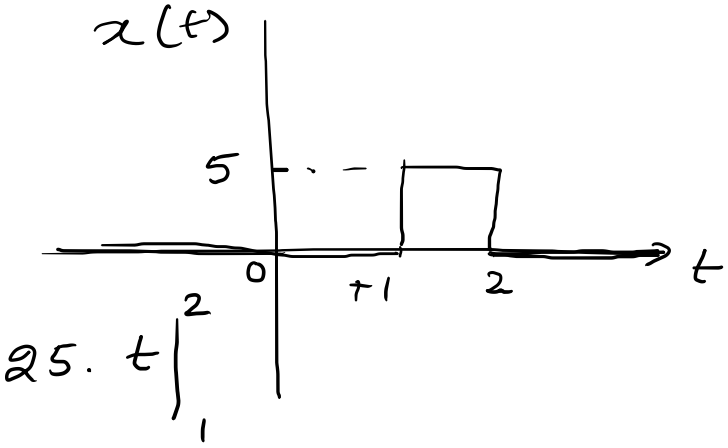
$$P_\infty = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 dt = \lim_{T\to\infty} \int_{-T}^{T} \frac{25\, dt}{2T}$$

$$= \lim_{T\to\infty} \frac{25}{2T} \cdot 2T = 25$$

P_∞ is finite because $x(t)$ is a power signal & non-zero.

Ex

$$x(t) = \begin{cases} 5, & 1 \le t \le 2 \\ 0, & \text{otherwise} \end{cases}$$



C.T. Signal
- Energy signal
- non-periodic signal or aperiodic signal

$$E_\infty = \int_{-\infty}^{\infty} |x(t)|^2 dt = 25 \int_{1}^{2} dt = 25.\, t\Big|_1^2$$

$$= 25(2-1) = 25$$

E_∞ = finite

$$P_\infty = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 \, dt$$

$$= \lim_{T\to\infty} \frac{1}{2T} \cdot 25 = 0$$

$$= \lim_{T\to\infty} \frac{E_\infty}{2T} = 0$$

$\therefore$ E_∞ is a finite non-zero constant value.

$$E_\infty = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{1} 0 \cdot dt + \int_{1}^{2} 25\, dt + \int_{2}^{\infty} 0 \cdot dt$$

($\int_{-\infty}^{1} 0 \cdot dt = 0$, $\int_{2}^{\infty} 0 \cdot dt = 0$)

$$= 25$$

Ex

$x(t) = t$

$\forall\ t$

$$P_\infty = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 dt$$

$$= \lim_{T\to\infty} \frac{1}{2T} \cdot \int_{0}^{T} t^2 dt$$

$$= \lim_{T\to\infty} \frac{1}{T} \cdot \frac{T^3}{3}$$

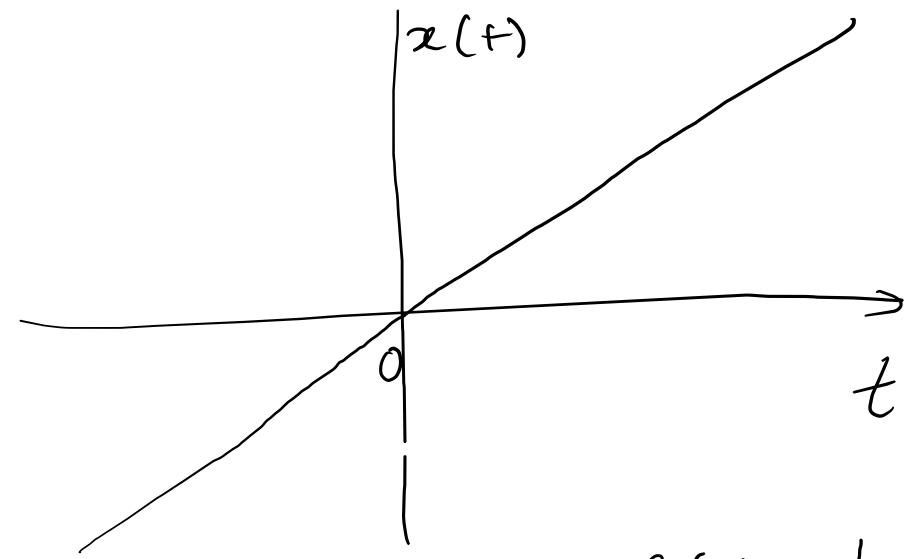
$$= \lim_{T\to\infty} \frac{T^2}{3} = \infty$$

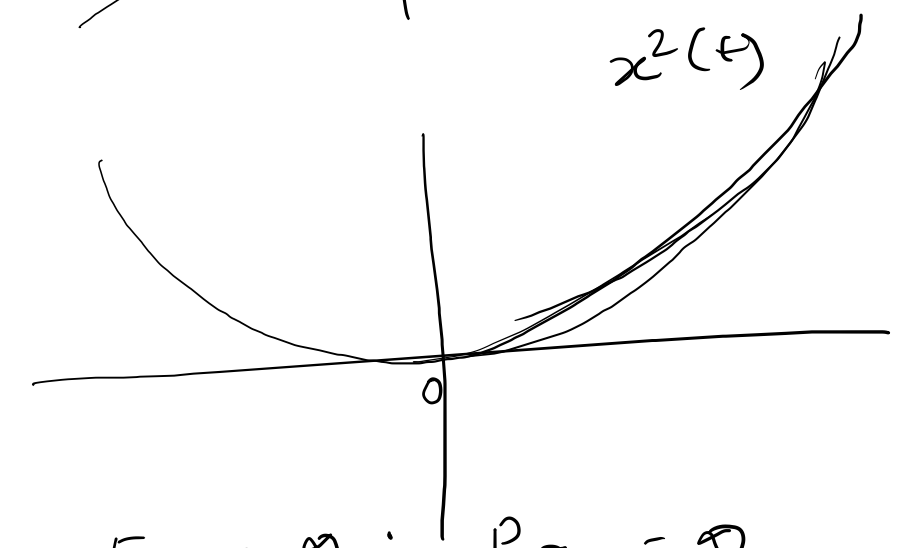
$$E_\infty = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$= \int_{-\infty}^{\infty} t^2 dt$$

$$= 2\int_{0}^{\infty} t^2 dt$$

$$= 2\,\frac{t^3}{3}\Big|_0^\infty = \infty$$





$E_\infty = \infty$; $P_\infty = 0$

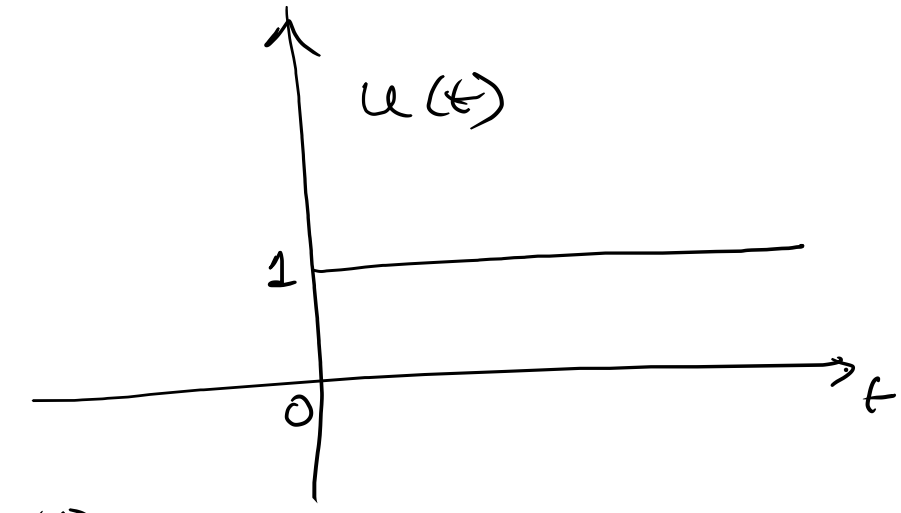
Neither an energy signal, nor a power signal.

Ex

$$x_1(t) = e^{-2t}\,u(t)$$

$u(t)$ ≡ unit-step function

$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & \text{otherwise} \end{cases}$$



$$x_1(0) = e^{0} \cdot 1 = 1$$

$$x_1(\infty) = e^{-\infty} \cdot 1 = 0 \cdot 1 = 0$$

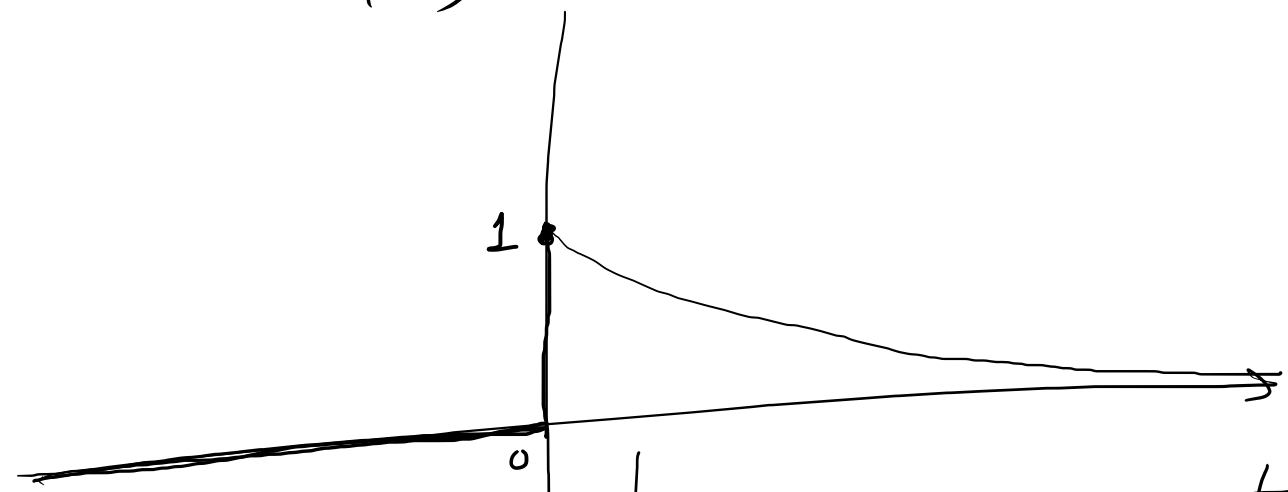
Find E_∞ & P_∞

$$E_\infty = \int_{-\infty}^{\infty} |x_1(t)|^2 dt = \int_{0}^{\infty} |e^{-2t}|^2 dt$$

$$= \int_{0}^{\infty} e^{-4t} dt = \frac{e^{-4t}}{-4}\Bigg|_0^\infty = \frac{0-1}{(-4)} = \frac{1}{4}$$

E_∞ ≡ finite non-zero

$x_1(t)$



$$P_\infty = \lim_{T\to\infty} \frac{E_\infty}{2T} = 0$$

$$x_2(t) = e^{j(2t+\pi/4)} \Rightarrow \text{periodic signal}$$

$$= e^{j2t} \cdot e^{j\pi/4}$$

$\equiv$ ↑ periodic

$$E_\infty = \int_{-\infty}^{\infty} |x_2(t)|^2 dt = \cdots \int_{-T/2}^{T/2} |x_2(t)|^2 dt + \int_{T/2}^{3T/2} |x_2(t)| dt$$

$$+ \cdots$$

$T \equiv$ time period

$$\int_{-T/2}^{T/2} |x(t)|^2 dt = \int_{-T/2}^{T/2} 1 \cdot dt = T$$

$$+T + \cdots$$

$$E_\infty = \cdots T + T + \cdots$$

$$= \infty$$

If $x(t) = x(t+T) \quad \forall T$

then $T \equiv$ Smallest positive no. for which the above eqn. holds true is called the fundamental time period.

$T =$ fundamental time period

$x_3(t) = G_0 t$

$\mathcal{E}_\infty = \int_{-\infty}^{\infty} |G_0(t)|^2 dt$

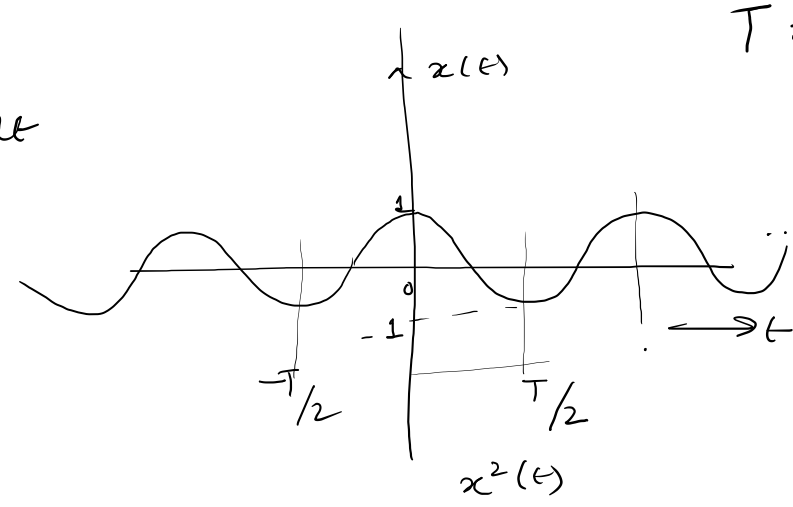
$E_T = \int_{-T/2}^{T/2} |G_0(t)|^2 dt$

$= \int_{-T/2}^{T/2} \frac{1+G_{02t}}{2} dt$

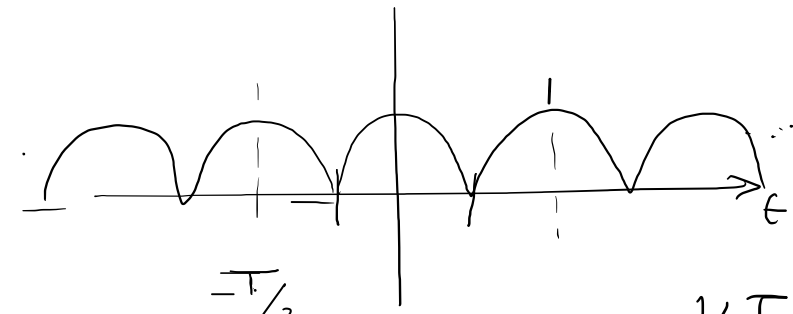
$= \frac{1}{2} \cdot T + \frac{1}{2} \int_{-T/2}^{T/2} G_{02t} dt$

$= \frac{1}{2} T$

$E_\infty = \lim_{T\to\infty} E_T = \infty$



$x^2(t)$



$P_\infty = \lim_{T\to\infty} \frac{E_T}{T} = \lim_{T\to\infty} \frac{\frac{1}{2}T}{T} = \lim_{T\to\infty} \frac{1}{2}$

$P_\infty = \frac{1}{2}$

Operations → Time shifting / time translation ...

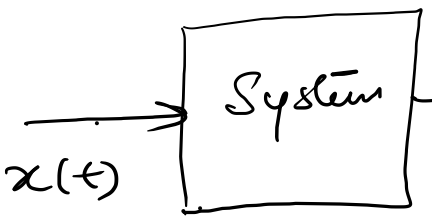
→ Time scaling

→ Time reversal

C.T. Signals

Time Shifting

$S:$ $\boxed{y(t) = x(t - t_0)}$ $\forall t$ where t_0 = constant



Let $t_0 = 3$

delayed the signal

$y(t) = x(t - 3)$

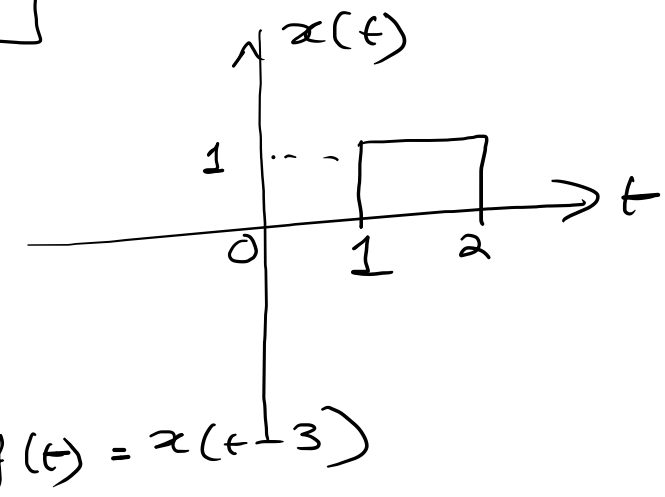
$y(0) = x(-3) = 0$

$y(1) = x(-2) = 0$

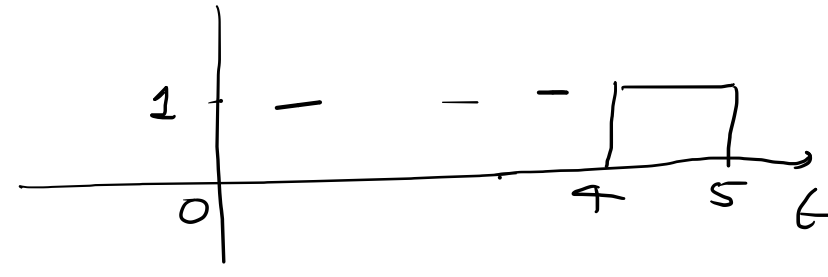
$y(2) = x(-1) = 0$

$y(3) = x(0) = 0$

$y(4) = x(1)$



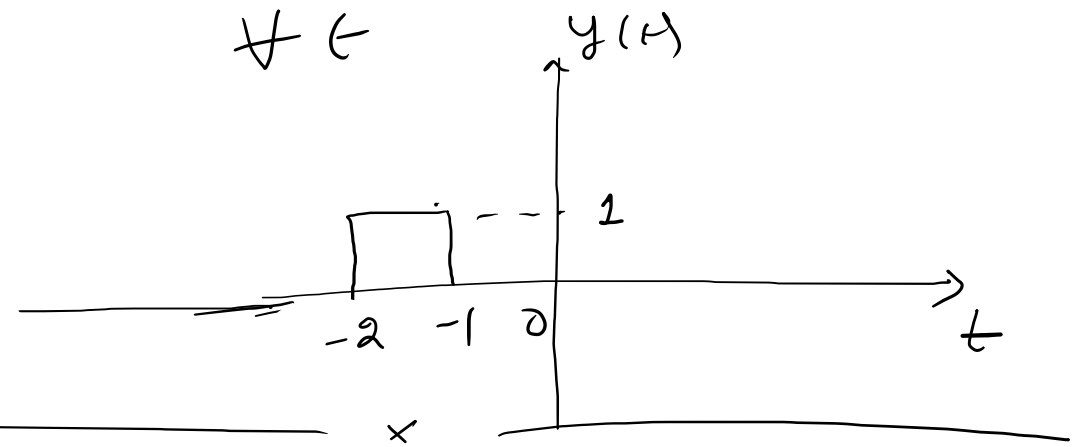
$y(t) = x(t - 3)$



Let $t_0 = -3$

$$y(t) = x(t+3)$$

Signal is advanced in time by 3 units



Time Scaling

where $a \equiv$ constant

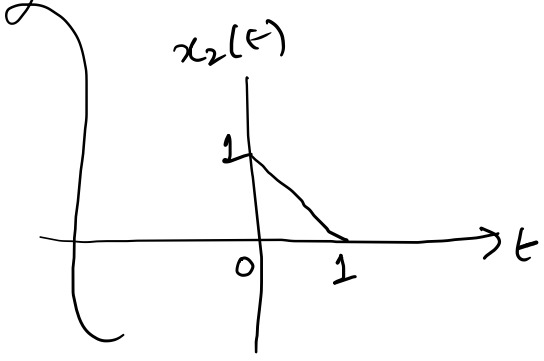
$$S_1 : y_1(t) = x(at)$$

Time reversal

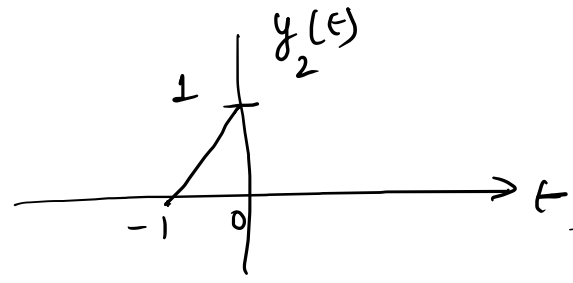
Ex-1

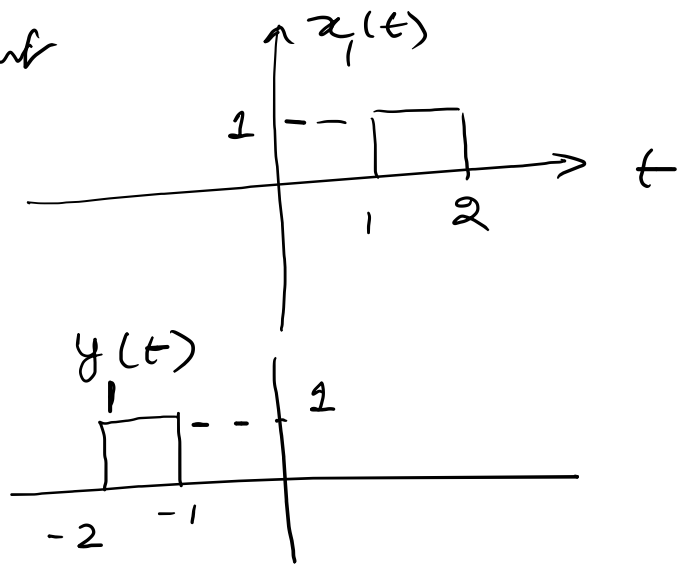
$a = -1$

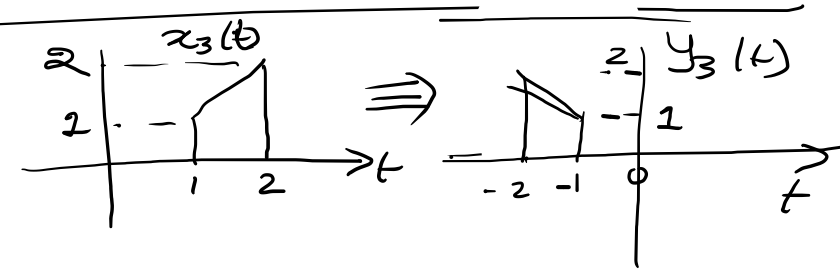
$$y_1(t) = x(-t)$$



$\Longrightarrow$



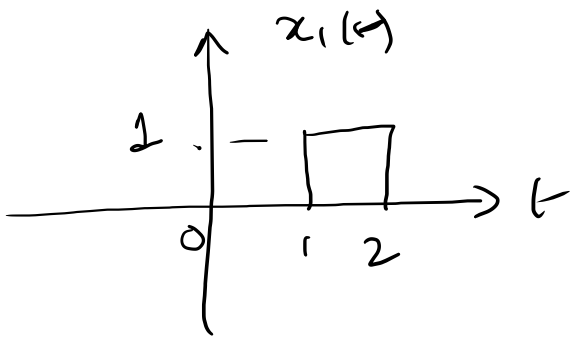


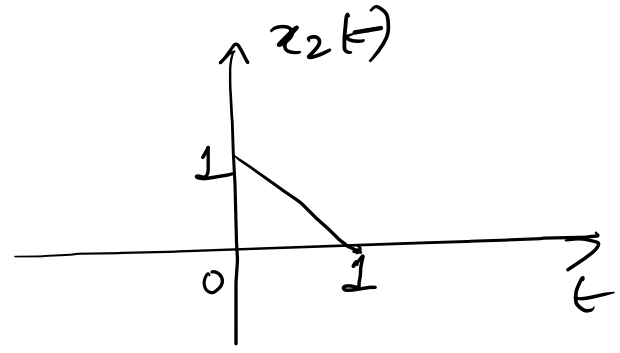


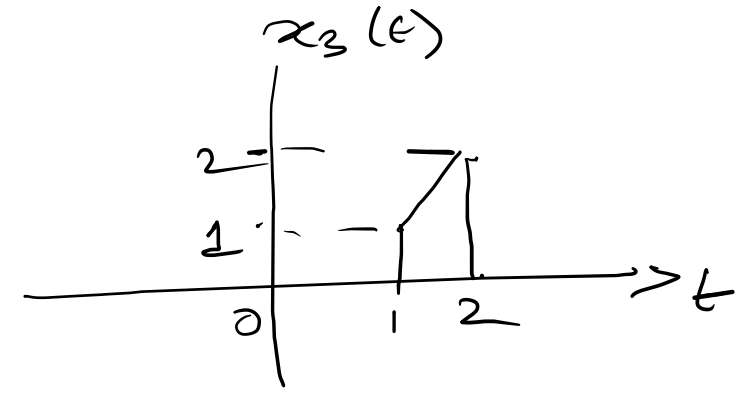
$a > 1$ Let $a = 2$

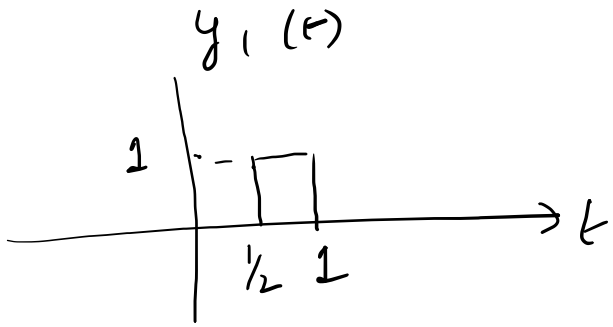
$y(t) = x(2t)$

playing a record at double the speed
compression of the signal in time



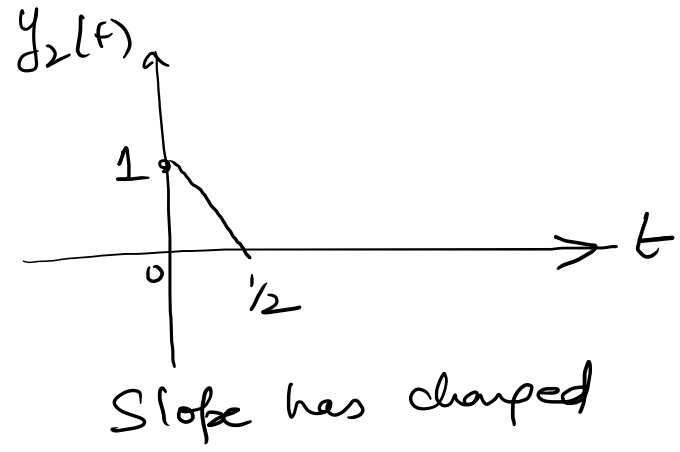




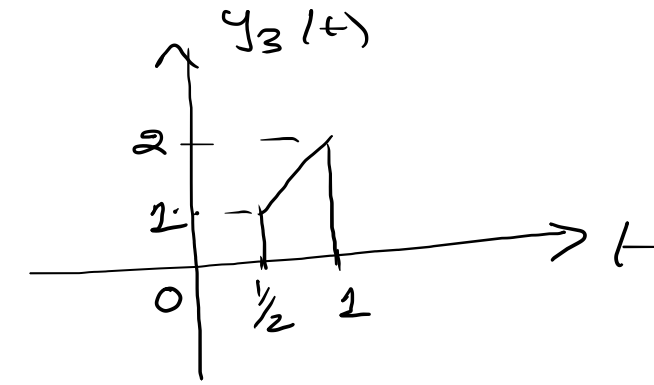


$y_1\left(\frac{1}{2}\right) = x\left(2 \cdot \frac{1}{2}\right) = x(1)$

$y_1(1) = x(2) = x(2)$



Slope has changed

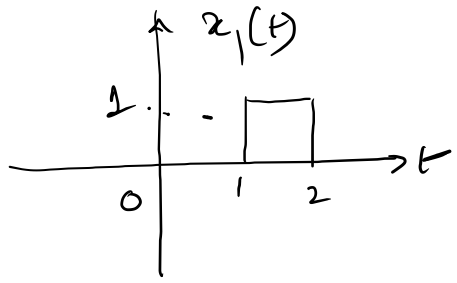


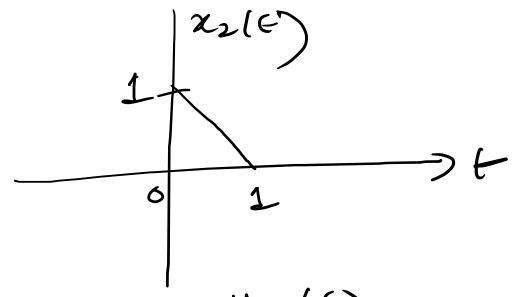
Slope has changed

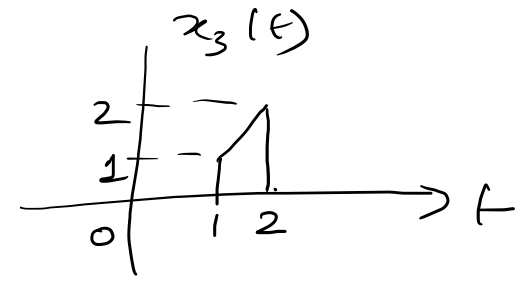
$0 < a < 1$

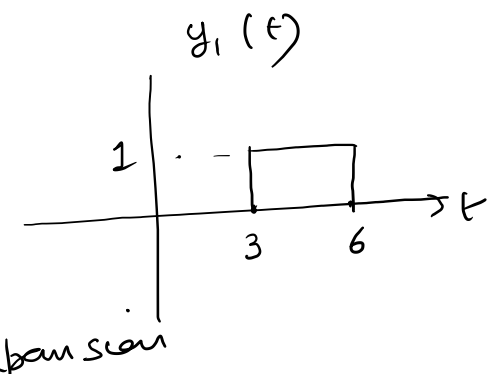
S: $y(t) = x(at)$ let $a = \frac{1}{3}$ | $y(t) = x$

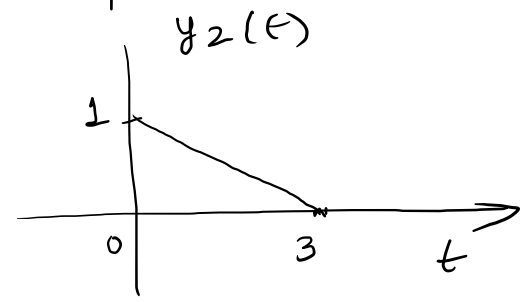
$y(t) = x\left(\frac{t}{3}\right)$

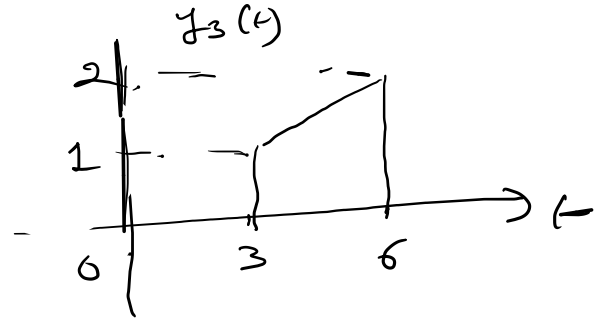












Time - expansion

$a = 1$ same signal $\Rightarrow$ Unit System

$a = -1$ time reversal

$a > 1$ time compression

$0 < a < 1$ time expansion

$a < -1$ time compression with reversal

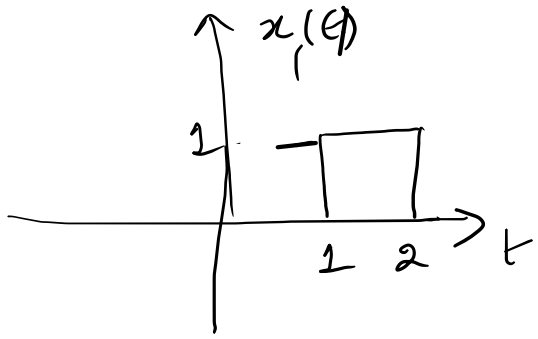
$-1 < a < 0$ time expansion with reversal

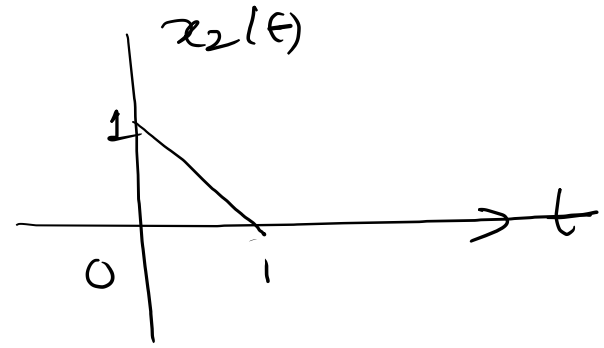
$y(t) = x(-2t)$

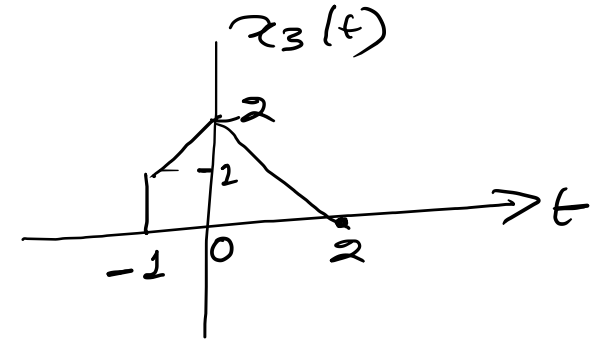
Time shifting with scaling

$$S: \quad y(t) = x(at + b) \qquad \forall\, t$$

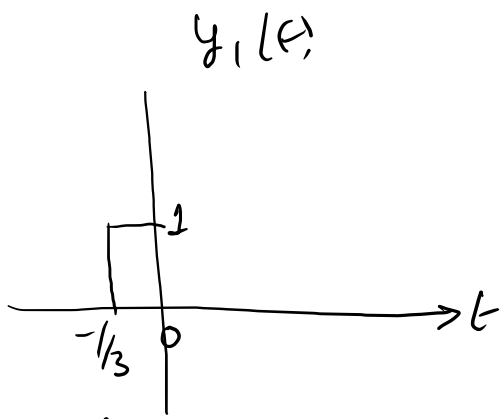
$$y(t) = x(3t+2)$$

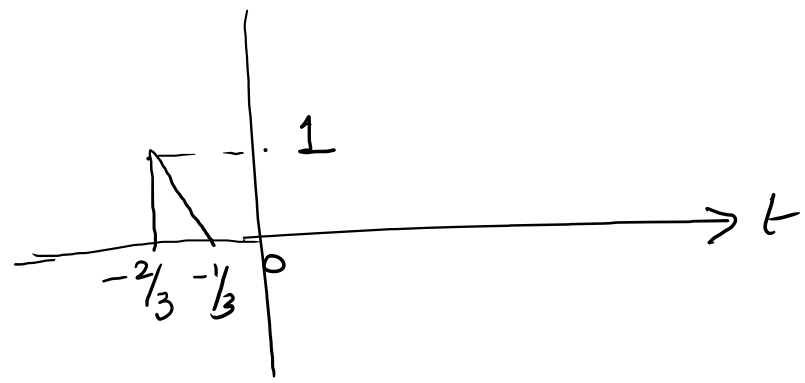


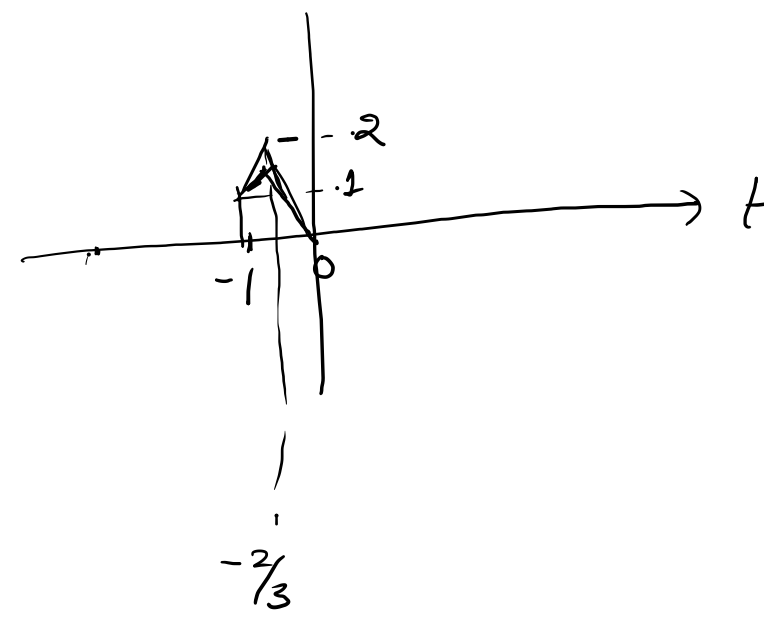




Hint:
Fixer shift
then
scale



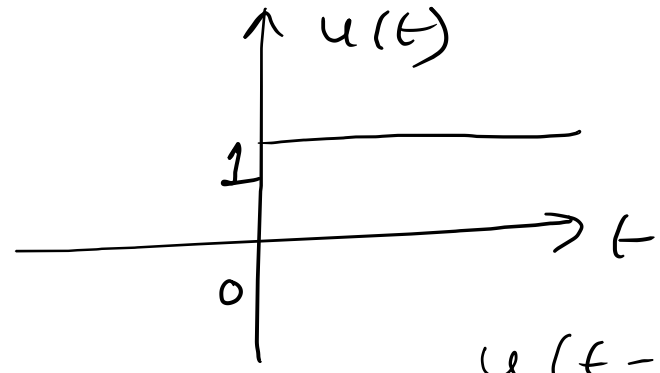




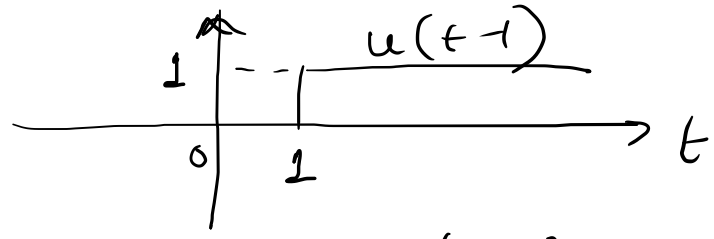
$$y\left(-\tfrac{1}{3}\right) = x(-1+2) = x(1)$$

$$y(0) = x(2)$$

$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & \text{otrosmie} \end{cases}$$

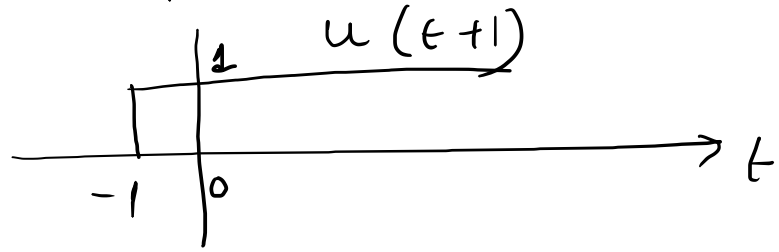


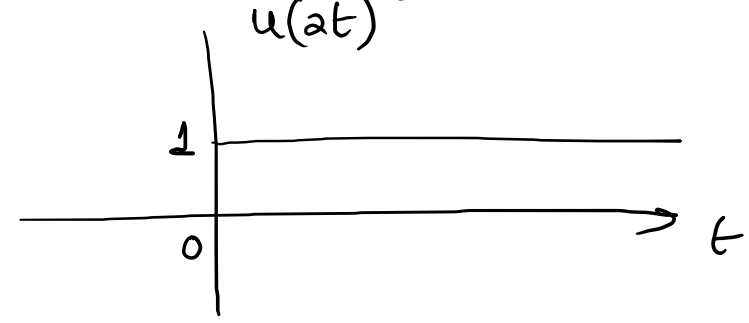
$u(t-1)$ $\Longrightarrow$



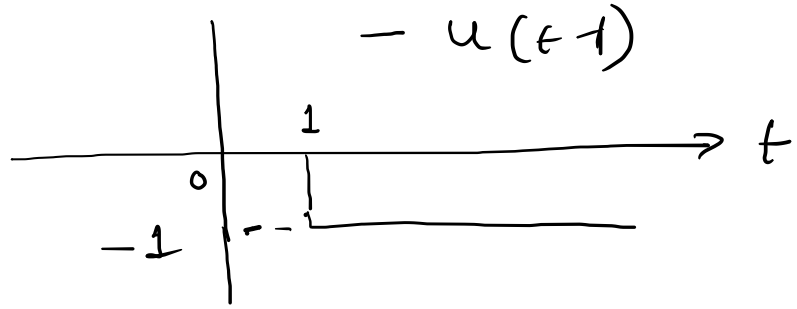
$$u(t-1) = \begin{cases} 1 & t \geq 1 \\ 0 & \text{otrosm.} \end{cases}$$

$u(t+1)$ $\Longrightarrow$

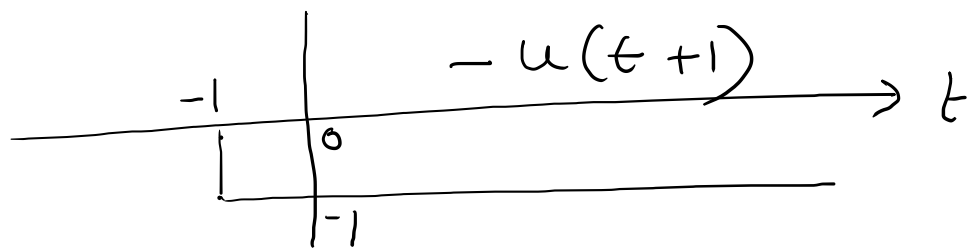




$-u(t-1)$

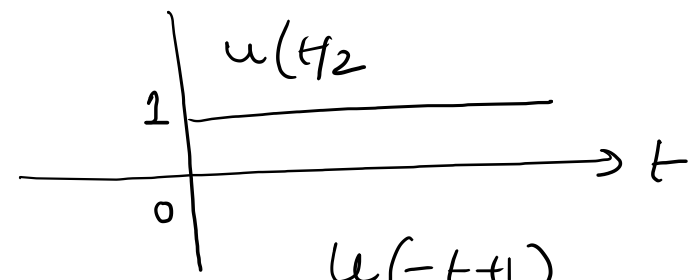


$-u(t+1)$

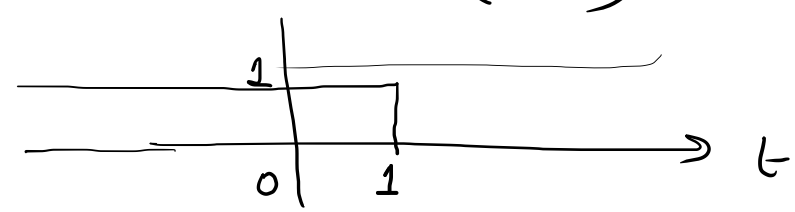


$u(2t)$

$u\left(\frac{t}{2}\right)$



$u(-t+1)$



$u(-t-1)$

